

APPENDIX A

all the math you
need... and no more
(an executive
summary)



The aim of this Appendix...

...is to give you an understanding of almost all the mathematics you need in quantitative finance, and to give you the confidence to read through the more technical parts of this and other books.

In this Appendix...

- e
- $\log$
- differentiation and Taylor series
- expectations and variances



A.1 INTRODUCTION

This book is for *everyone* interested in quantitative finance. This subject is increasingly technical. Some people don't have a high-level math training but may still be interested in the technical side of things. In this chapter we look at the mathematics that you need to cope with the vast majority of derivatives theory and practice. Although often couched in very high

level mathematics almost all finance theory can be interpreted using only the basics that I describe here.

The most useful math is the simplest math. This is particularly true in finance where the beauty of the mathematics can and does lead to people not seeing the wood for the trees. All basic finance theory and a great deal of the advanced research really require only elementary mathematics if approached in the right way. In this chapter I explain this elementary mathematics. I am trying to do for mathematical finance what Seuss (1999) did for the English language.



A simple, ubiquitous function

A.2 e

The first bit of math you need to know about is e .
 e is

- a number, 2.7183 ...
- a function when written e^x , this function is a.k.a. $\exp(x)$

The function e^x is just the number 2.7183... raised to the power x ; e^2 is just $2.7183...^2 = 7.3891...$, e^1 is 2.7183... and $e^0 = 1$. What about non-integer powers?

The function e^x can be written as an infinite series

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots = \sum_{i=0}^{\infty} \frac{x^i}{i!}.$$

This gets around the non-integer power problem.

A plot of e^x as a function of x is shown in Figure A.1.

The function e^x has the special property that the slope or gradient of the function is also e^x . Plot this slope as a function of x and for e^x you get the same curve again. It follows that the slope of the slope is also e^x , etc.



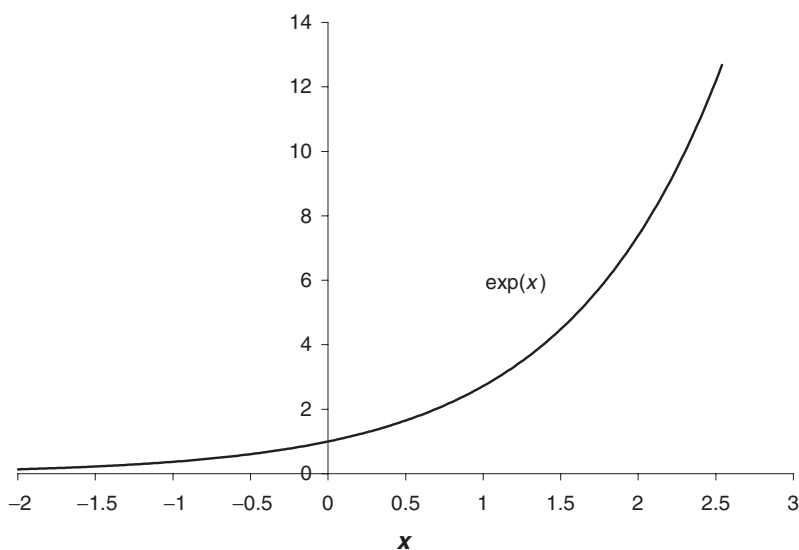
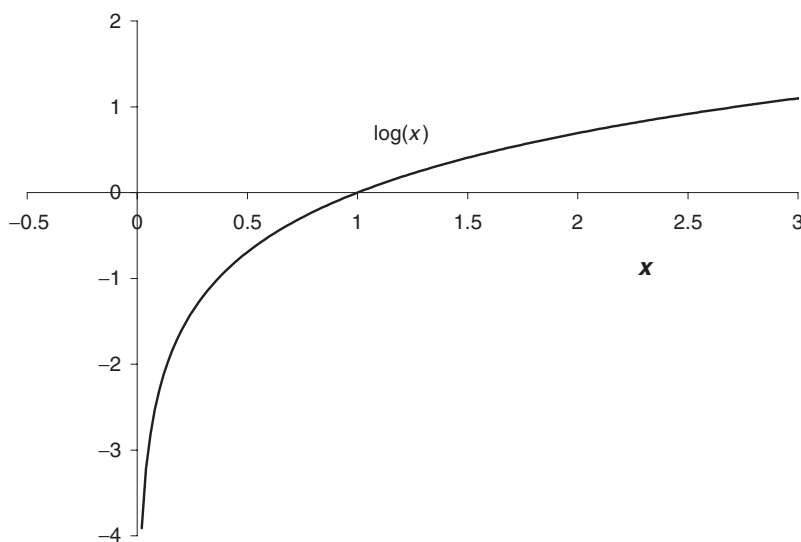
The inverse of the exponential

A.3 $\log$

Take the plot of e^x , Figure A.1, and rotate it about a 45° line to get Figure A.2. This new function is $\ln x$, the natural logarithm of x . The relationship between $\ln$ and e is

$$e^{\ln x} = x \quad \text{or} \quad \ln(e^x) = x.$$

So, in a sense, they are inverses of each other.

**Figure A.1** The function e^x .**Figure A.2** The function $\log x$.

The function $\ln x$ is also often denoted by $\log x$, as in this book. Sometimes $\log x$ refers to the function with the properties

$$10^{\log x} = x \quad \text{and} \quad \log(10^x) = x.$$

This function would be called ‘logarithm base 10.’ The most useful logarithm has base $e = 2.7183\dots$ because of the properties of the gradient of e^x .

The slope of the $\log x$ function is x^{-1} .

From the figure you can see that there don't appear to be any values for $\log x$ for negative x . The function can be defined for these but you'd need to know about complex numbers, something we won't be requiring here.

A.4 DIFFERENTIATION AND TAYLOR SERIES

I've introduced the idea of a gradient or slope in the section above. If we have a function denoted by $f(x)$, then we denote the gradient of this function at the point x by

$$\frac{df}{dx}.$$

Mathematically the slope is defined as

$$\frac{df}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}.$$

The action of finding the gradient is also called 'differentiating' and the slope can also be called the 'derivative' of the function. This use of 'derivative' shouldn't be confused with the use meaning an option contract.



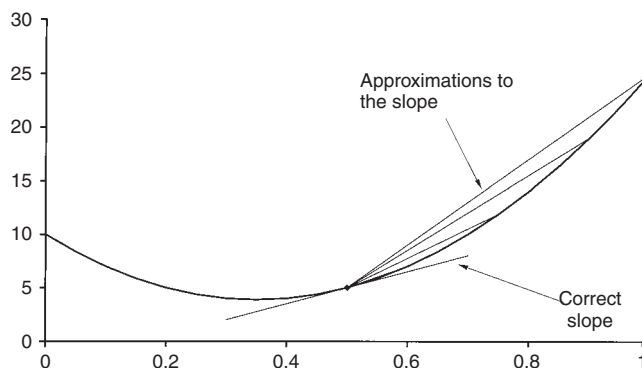
Time Out...

Differentiation

As I've said before, the slope is exactly the same as the gradient you experience in your car, and represented by signs such as '1-in-10 gradient.' Look at the figure below to see how the limit δx tending to zero might work in practice.



Gradients



The slope can also be differentiated, resulting in a second derivative of the function $f(x)$. This is denoted by

$$\frac{d^2f}{dx^2}.$$

We can take this differentiation to higher and higher orders.

Take a look at Figure A.3. In particular, note the two dots marked on the bold curve. The bold curve is the function $f(x)$. The dot on the left is at the point x on the horizontal axis and the function value is $f(x)$, the distance up the vertical axis. The dot to the right of this is at $x + \delta x$ with function value $f(x + \delta x)$. What can we say about the vertical distance between the two dots in terms of the horizontal distance?

Start with a trivial example. If the distance δx is zero then the vertical distance is also zero. Now consider a very small but non-zero δx .

The straight line tangential to the bold curve $f(x)$ at the point x is shown in the figure. This line has slope $\frac{df}{dx}$ evaluated at x . Notice that the right-hand hollow dot is almost on this bold line. This suggests that a good approximation to the value $f(x + \delta x)$ is

$$f(x + \delta x) \approx f(x) + \delta x \frac{df}{dx}(x).$$



Gradients of gradients

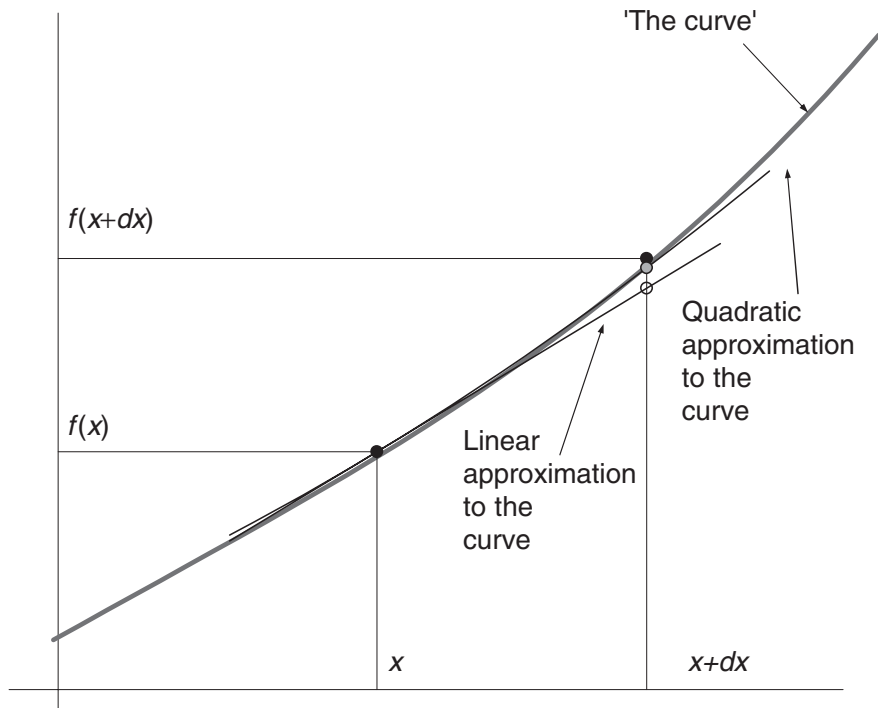


Figure A.3 A schematic diagram of Taylor series.

This is a linear relationship between $f(x + \delta x) - f(x)$ and δx . This makes sense since on rearranging we get

$$\frac{df}{dx} \approx \frac{f(x + \delta x) - f(x)}{\delta x}$$

which as δx goes to zero becomes our earlier definition of the gradient.



But the right-hand hollow dot is not *exactly* on the straight line. It is slightly above it. Perhaps a quadratic relationship between $f(x + \delta x) - f(x)$ and δx would be a more accurate approximation. This is indeed true (provided δx is small enough) and we can write

$$f(x + \delta x) \approx f(x) + \delta x \frac{df}{dx}(x) + \frac{1}{2} \delta x^2 \frac{d^2 f}{dx^2}(x).$$

This approximation, shown on the figure as the gray dot, is more accurate. One can take this approximation to cubic, quartic, The Taylor series representation of $f(x + \delta x)$ is the infinite sum

See how accurate
this can be

$$f(x + \delta x) = f(x) + \sum_{i=1}^{\infty} \frac{1}{i!} \delta x^i \frac{d^i f}{dx^i}(x).$$

Taylor series is incredibly useful in derivatives theory. In derivatives theory the function that we are interested in, instead of being f , is V , the value of an option. The independent variable is no longer x but is S , the price of the underlying asset. From day to day the asset price changes by a small, random amount. This asset price change is just δS (instead of δx). The first derivative of the option value with respect to the asset is known as the delta, and the second derivative is the gamma.

The value of an option is not only a function of the asset price S but also the time t : $V(S, t)$. This brings us into the world of partial differentiation.

Think of the function $V(S, t)$ as a surface with coordinates S and t on a horizontal plane. The function V is the height of a hill above sea level with S and t being distances, or map coordinates, in the northerly and westerly directions. The *partial* derivative of $V(S, t)$ with respect to S is written

$$\frac{\partial V}{\partial S}$$

and is defined as

$$\frac{\partial V}{\partial S} = \lim_{\delta S \rightarrow 0} \frac{V(S + \delta S, t) - V(S, t)}{\delta S}.$$

Note that in this V is only ever evaluated at time t . This is like measuring the gradient of the function $V(S, t)$ in the S direction along a constant value of t , i.e. the slope of our hillside in the northerly direction. Note also that we are now using a special curly ∂ instead of the roman d . The partial derivative of $V(S, t)$ with respect to t is similarly defined as

$$\frac{\partial V}{\partial t} = \lim_{\delta t \rightarrow 0} \frac{V(S, t + \delta t) - V(S, t)}{\delta t}$$

and is the slope of the hill in the westerly direction.

Higher-order derivatives are defined in the obvious manner.
The Taylor series expansion of the value of an option is then

$$V(S + \delta S, t + \delta t) \approx V(S, t) + \delta t \frac{\partial V}{\partial t} + \delta S \frac{\partial V}{\partial S} + \frac{1}{2} \delta S^2 \frac{\partial^2 V}{\partial S^2} + \dots$$

This series goes on for ever, but I've only written down the largest and most important terms, those which are required for the Black–Scholes analysis.

A.5 DIFFERENTIAL EQUATIONS

Right at the beginning of the book we saw a differential equation. Remember the equation for money in the bank

$$\frac{dM}{dt} = rM?$$

This is an example of an **ordinary differential equation**. ‘Ordinary’ because it involves only derivatives with respect to a single variable; there are no curly ∂ s. This equation can be solved to find M as a function of the other variable t .

Throughout this book there are **partial differential equations**. These involve the curly ∂ s because there are derivatives/slopes/gradients with respect to more than one variable. The Black–Scholes equation is an example. In our mountain example, think of a partial differential equation as relating the height of the mountain to its slope in each direction, and maybe even the slope of the slope. Again, such equations must be solved to find the height of the mountain, the option value. How this is done analytically (if possible) or numerically (always possible) will be discussed later.

A.6 MEAN, STANDARD DEVIATION AND DISTRIBUTIONS

Much of the modeling in finance uses ideas from probability theory. Again you don't need to know that much to understand most of quantitative finance.

The first idea is that of the mean. The mean is the same as the average or the expectation. What is the average shoe size of US males?

If you roll a die there is an equal, $\frac{1}{6}$, probability of each number coming up. What is the expected number or the average number if you roll the die many times. The answer is

$$1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = 3\frac{1}{2}.$$

Here we just multiply each of the possible numbers that could turn up by the probability of each, and sum. Although $3\frac{1}{2}$ is the expected value, it cannot, of course, be thrown since only integers are possible.

Generally, if we have a random variable X (the number thrown, say) which can take any of the values x_i (1, 2, 3, 4, 5, 6 in our example) for $i = 1, \dots, N$ each of which has a probability $P(X = x_i)$ (in the example, $\frac{1}{6}$) then the expected value is

$$E[X] = \sum_{i=1}^N x_i P(X = x_i).$$

Expectations have the following properties:

$$E[cX] = cE[X]$$

and

$$E[X + Y] = E[X] + E[Y].$$

If the outcome of two random events X and Y have no impact on each other they are said to be independent. If X and Y are independent we have

$$E[XY] = E[X]E[Y].$$

Expectations are important in finance because we often want to know what we can expect to make from an investment on average.

The expectation or mean is also known as the first moment of the distribution of the random variable X . It can be thought of as being a typical value for X . The scatter of values around the mean can be measured by the second moment or the variance:

$$\text{Var}(X) = E[(X - E[X])^2].$$

Variances have the following property:

$$\text{Var}(cX) = c^2 \text{Var}(X).$$

When X and Y are independent

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

The standard deviation is the square root of the variance and is perhaps more useful as a measure of dispersion since it has the same units as the variable X :

$$\text{Standard deviation}(X) = \sqrt{\text{Var}(X)}.$$

The standard deviation is just a measure of the spread around that mean. The smaller the standard deviation of shoe sizes, the less variation in sizes among the population. If the standard deviation is large, there is a great deal of variation in shoe sizes.

Standard deviations are important in finance because they are often used as a measure of risk in an investment. The higher the standard deviation of investment returns the greater the dispersion of the returns and the greater the risk.

If we were to plot a bar chart (histogram) of number of the population with each shoe size on the vertical axis and shoe size on the horizontal we'd have a frequency distribution of shoe sizes. Now scale the height of each bar with the total number in the population and you've got a probability density function. Why did I ask you to scale the distribution? So that the heights of all the bars add up to one, the probability of having *some* shoe size is one.

That was an example of a discrete distribution. Shoe sizes are a discrete set. Contrast that with foot length. The length of a foot can



Some examples

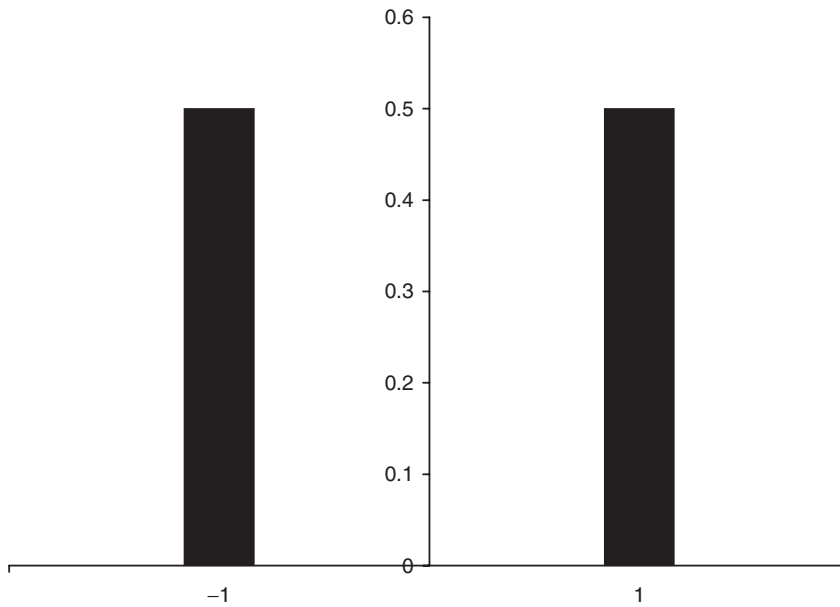


Figure A.4 Probability density functions for tossing a fair coin; a head wins \$1 a tail loses \$1. The mean is zero.

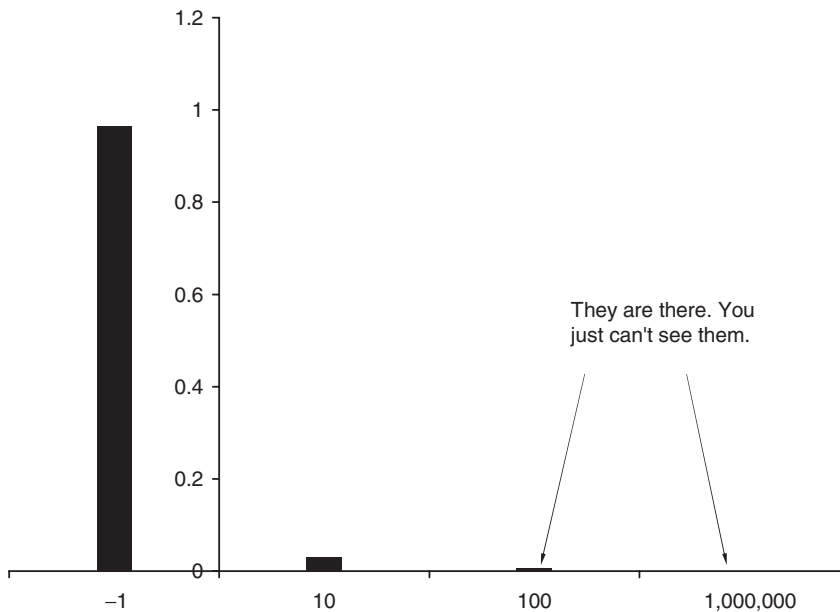


Figure A.5 Probability density function for a lottery. The mean is negative.

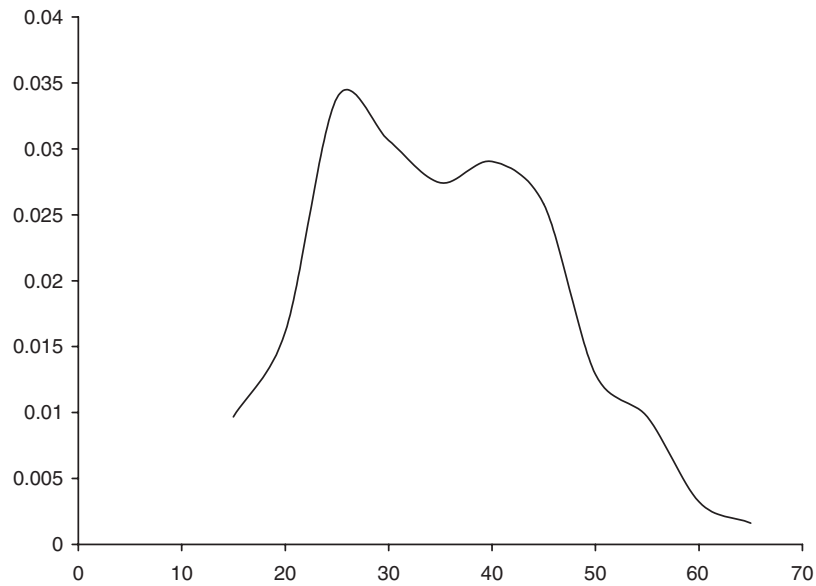


Figure A.6 Probability density function for the amount I spend on Amazon at 3a.m. when I can't sleep. A continuous distribution.

be any number, not confined to a discrete set. Foot lengths form a continuous distribution. In practice, given a continuous distribution such as foot length you will have to assign each length to a finite sized 'bucket.' Think of the bucket as being the corresponding shoe size.

Above are a few examples of probability density functions.

A.7 **SUMMARY**

The math you need in quantitative finance is surprisingly little. And you've just seen all of it.